Project 2

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November 2024

1.1 2pBVP solvers

Consider the bounduary value probelm:

$$y''(x) = -(\pi^2)sin(\pi x),\ \ y(0) = 0, y(1) = 1,$$

It has exact solution:

$$y(x) = sin(\pi x) + x.$$

- Grid: $x_k = k\Delta x$, $\Delta x = \frac{1}{N+1}$

Discretized problem:

$$\frac{y_{i+1} - 2y_i + y_{i-1}}{\Delta x^2} = -(\pi^2)sin(\pi x_i),\ \ y_0 = 0, y_{N+1} = 1.$$

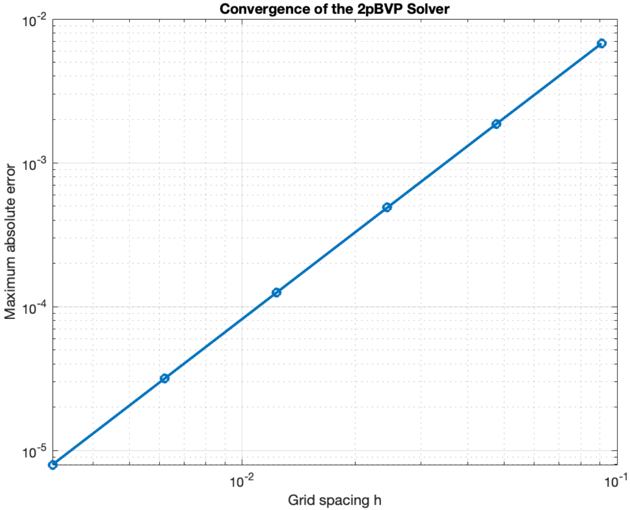
The right hand of the system of the equations we have will be:

$$[f(x_1) - \frac{\alpha}{\Delta x^2}, f(x_2), ..., f(x_N) - \frac{\beta}{\Delta x^2}]^t, f(x_i) = -(\pi^2)sin(\pi x_i).$$

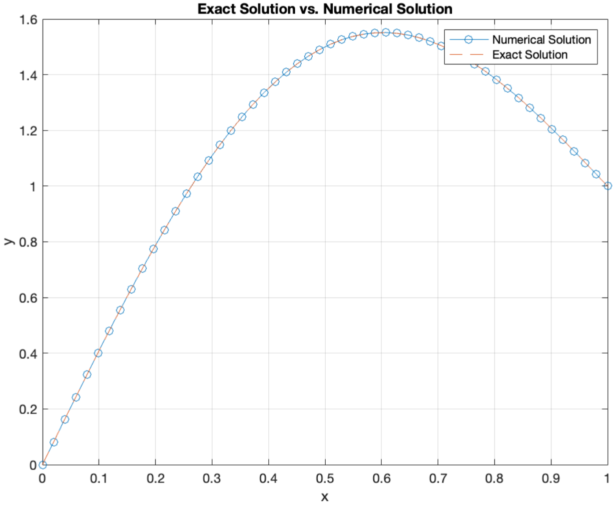
1.1

```
function y=twopBVP(fvec, alpha, beta, L, N)

dx=L/(N+1);

%constructing the tridiagonal matrix following hint 4 given

main=-2*ones(N,1);
sup=ones(N-1,1);
sub=ones(N-1,1);

A=diag(sub,-1)+diag(sup,1)+diag(main,0); %from hint 4

% scale matrix by dx^2
A = A / dx^2;

%adjust the matrix to correct the first and last row of the matrix

rhs = fvec(:); %there was an error because matrix dimension
if length(rhs) ~= N
        error('rhs vector must be of size %d, but got %d', N, length(rhs));
    end
rhs(1)=rhs(1)-alpha/dx^2; %dx^2 is delta x^2, this will include alpha in th
rhs(end)=rhs(end)-beta/dx^2; %this will include beta in the last eq

%solve the linear system
y_int=A\rhs;
```

1.1 Second Order Convergence



1.1 Plotting the exact solution vs numerical solution



1.2 The Beam Equation

$$M'' = q(x), u'' = \frac{M(x)}{EI};$$

- $q(x) =$ load density (N/m)
- $M(x)$ is the bending moment (Nm);
- E is Young's modulus of elasticity (N/m^2).
- I the beam's cross-section moment of inertia (m^4)
- u is the beam's centerline deflection (m).
- $u(0) = u(L) = 0$
- $M(0) = M(L) = 0.$

1.2

- N=999 interior grid points
- Solving first for the bending moment M
- Solving for the deflection u

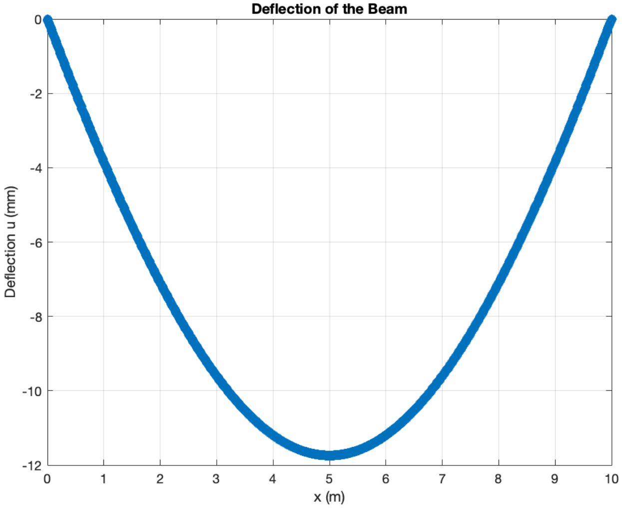
```
% solve for bending moment M(x)
fvec_M = q(x(:)); % right-hand side for M''(x) = q(x
fvec_M = fvec_M(:); %to ensure it's a column vector,
alpha_M = 0; % Boundary condition M(0) = 0
beta_M = 0; % Boundary condition M(L) = 0
M = twopBVP(fvec_M, alpha_M, beta_M, L, N); %solving
```

Figure: Bending Moment

```
fvec_u = M_interior ./ (E * I_vals);
fvec_u = fvec_u(:);
alpha_u = 0; % Boundary condition u(0) = 0
beta_u = 0; % Boundary condition u(L) = 0
u = twopBVP(fvec_u, alpha_u, beta_u, L, N);
```

Figure: Deflection

1.2 The Beam Equation



1.2

Deflection at midpoint : $-11.741059mm$

- Deflection curve is symmetric about the midpoint. (Load density $q(x)$ is uniform, i.e, uniform loading and the boundary conditions are symmetric)
- The boundaries are pinned at 0 and L and has deflection 0 which basically shows that the beam ends are fixed.
- The maximum deflection occurs at the midpoint of the beam. This shows how much the beam bends at its weakest point under the given load and stiffness conditions.

Part 2. Sturm Liouville eigenvalue problems

- Consider the boundary value problem

$$u''(x) = \lambda u(x), \quad u(0) = 0, u'(1) = 0.$$

  Using Approach 2:
- Grid: $x_k = k\Delta x$
- $\Delta x = \frac{1}{N+\frac{1}{2}}$
- Approximate u'' by $\frac{u_{n-1}-2u_n+u_{n+1}}{\Delta x^2}$, $n = 1, ..., N-1$ and the boundary conditions by $u_0 = 0$ and $\frac{u_{N+1}-u_N}{\Delta x} = 0 \iff u_{N+1} = u_N$.

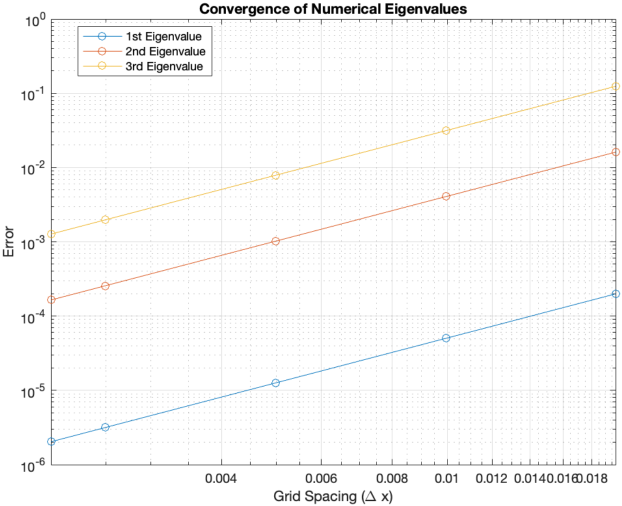
```
function [eigenvalues, eigenmodes, x] = sturm_liouville_solver(N)
    % inputs:
    %   N: number of interior grid points
    % outputs:
    %   eigenvalues
    %   eigenmodes
    %   x: grid including boundary points
    % Domain and grid
    L = 1;
    format long g
    dx = L / (N + (1/2)); % second order approach
    x = linspace(0, L, N + 2)'; % Include boundary points
    % Tridiagonal matrix for T_dx
    beta = 0; %for the neumann condition
    main_diag = (-2/dx^2) * ones(N, 1);
    off_diag = (1/dx^2)*ones(N-1, 1); %
    T_dx = diag(main_diag) + diag(off_diag, -1) + diag(off_diag, 1);
    T_dx(N, N - 1) = 1 / dx^2;
    T_dx(N, N) =  -1 / dx^2; % adjust diagonal entry for the last row
```

2.1

With approach 2:

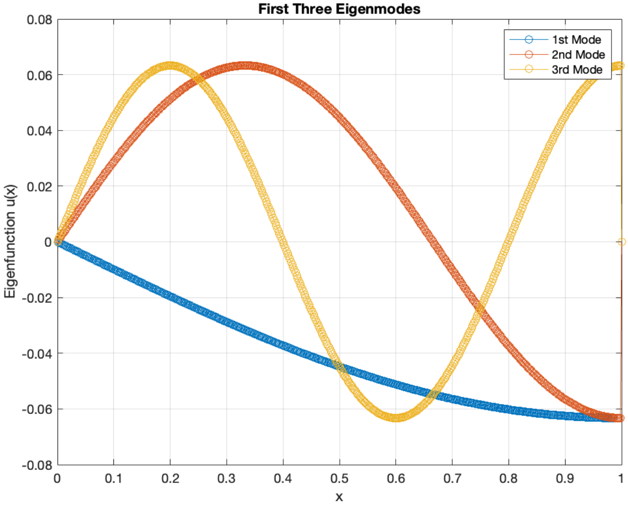
- N=499
- Eigenvalues numerical:
  $\lambda_1 : -2.46739907, \lambda_2 : -22.20644520, \lambda_3 : -61.68375663.$
- Eigenvalues analytical:
  $\lambda_1 = -2.46740110, \lambda_2 = -22.20660990, \lambda_3 = -61.68502751$
- $u_n = C_n sin(\omega_n x)$, with $\omega_n^2 = \lambda_n$.
- Second order convergences for the eigenvalues

2.1



Convergence of Numerical Eigenvalues
1st Eigenvalue
2nd Eigenvalue
3rd Eigenvalue
Error
Grid Spacing (Δ x)
0.004
0.006
0.008
0.01
0.012
0.014
0.016
0.018

2.1


First Three Eigenmodes
1st Mode
2nd Mode
3rd Mode
Eigenfunction u(x)
x
0.08
0.06
0.04
0.02
0
-0.02
-0.04
-0.06
-0.08
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1

2.2 Schrödinger equation

Sturm - Liouville problem for determining the wave function, energy levels and probability densities $|\psi|^2$ to find the probability of finding the particle in a certain position x.

- Schrödinger equation: $\psi'' - V(x)\psi = -E\psi$
- Boundary conditions: $\psi(0) = \psi(1) = 0$
- Wave function $\psi(x)$
- One-dimensional potential $V(x) \geq 0$
- Energy level E

2.2 $V(x) = 0$

Case $V(x) = 0$:

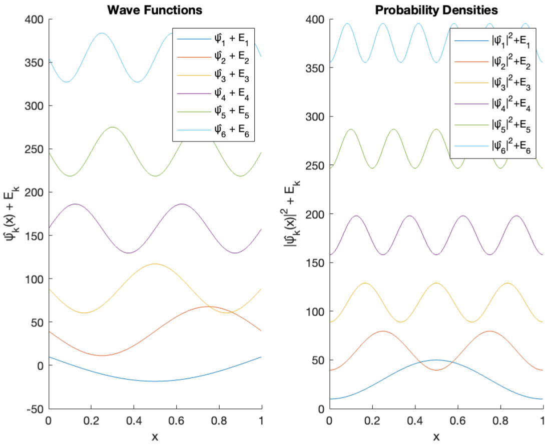
- $\psi'' = -E\psi$, $\psi(0) = \psi(1) = 0$.
- Grid: $x_k = k\Delta x$, $\Delta x = \frac{1}{N+1}$
- Discretized problem:

$$\frac{\psi_{i+1} - 2\psi_i + \psi_{i-1}}{\Delta x^2} = -E\psi_i, \psi_0 = \psi_{N+1} = 0.$$

2.2

```
function [eigenvalues, eigenvectors] = schrodinger_solver(V, x)

    % input:
    %   V - potential
    %   x - position
    % output:
    %   eigenvalues - energy levels
    %   eigenvectors -  wave functions

    N = length(x) - 2; % number of interior points
    dx = x(2) - x(1);  % grid spacing, it's the difference between two consecutive points

    %main diagonal: 2/dx^2 + V(x) (just interior points)
    V=V(2:end-1)';
    main_diag = 2 / dx^2 + V;

    % off-diagonal: -1/dx^2
    off_diag = -1 / dx^2 * ones(N-1, 1);

    % tridiagonal matrix
    T = diag(main_diag) + diag(off_diag, 1) + diag(off_diag, -1);
```

2.2 V(x)=0


Wave Functions
Probability Densities
ψ̂_1 + E_1
ψ̂_2 + E_2
ψ̂_3 + E_3
ψ̂_4 + E_4
ψ̂_5 + E_5
ψ̂_6 + E_6
|ψ̂_1|^2+E_1
|ψ̂_2|^2+E_2
|ψ̂_3|^2+E_3
|ψ̂_4|^2+E_4
|ψ̂_5|^2+E_5
|ψ̂_6|^2+E_6
ψ̂_k(x) + E_k
|ψ̂_k(x)|^2 + E_k
x

2.2 V(x)=0

- The eigenvalues analytical are: $E_k = k^2\pi^2$ $w_k^2 = E_k$ $w_k = k\pi$
- The eigenfunctions (wave functions): $\psi_k = C_k sin(w_k x)$.
- The first six numerical eigenvalues are:

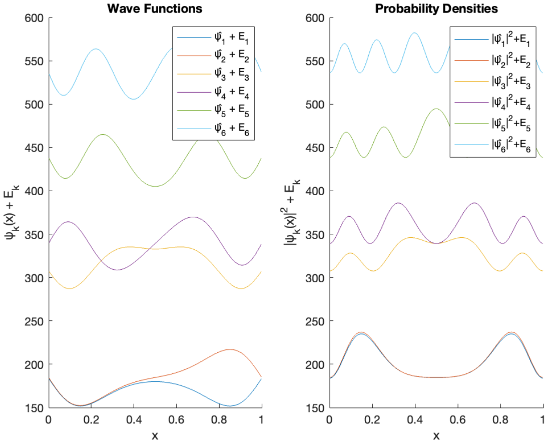
$$\lambda_1 = 9.8696, \lambda_2 = 39.4779, \lambda_3 = 88.8238, \lambda_4 = 157.9054,$$

$\lambda_5 = 246.7199, \lambda_6 = 355.2638.$

2.2 Potential Barriers

- $V(x) = 700(0.5 - |x - 0.5|)$

Doublet state: two states of almost the same energy but different parity due to the potential barrier inside the box (it can be seen for low levels).

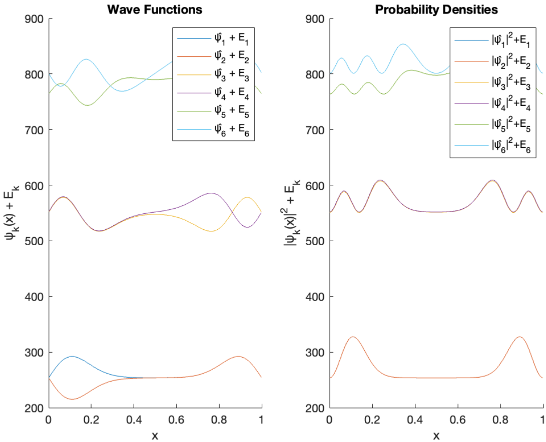


2.2 x=0.5

Energy levels where the probability of finding the particle at $x = 0.5$ becomes significant. The higher is for E_5, but also E_3.

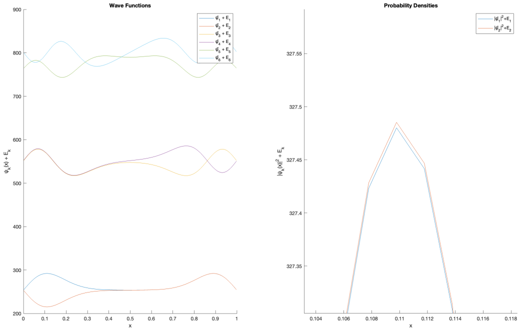
- $V(x) = 800sin^2\pi x$

Doublet state: two states of almost the same energy but different parity due to the potential barrier inside the box (it can be seen for low levels).



- $V(x) = 800 sin^2 \pi x$

Doublet state: two states of almost the same energy but different parity due to the potential barrier inside the box (it can be seen for low levels).



2.2

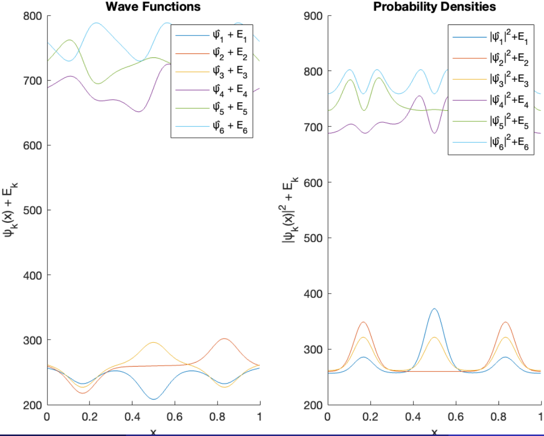
- $V(x) = 800 sin^2 \pi x$

The probability to find the particle at $x = 0.5$ becomes significant at E_5.

2.2

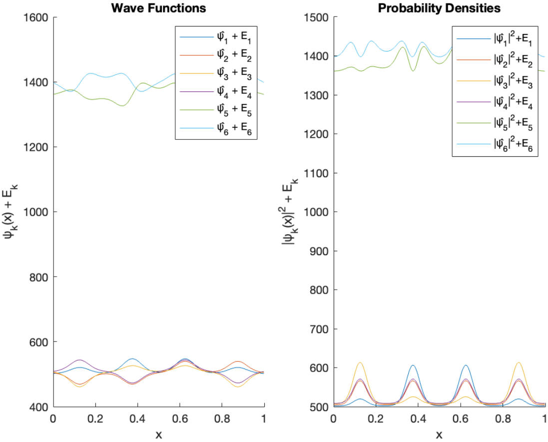
- $V(x) = 900cos^2(3\pi x)$

Potential with three wells so we have triplet state, i.e. three states of almost the same energy but different parity due to the potential barrier inside the box. We can see it in the probability densities plot.



2.2 Case up to us

- $V(x) = 1900cos^2(4\pi x)$



2.2 Case up to us

We can see in the probability densities plot that we have 4 wells.

- $V(x) = 1900cos^2(4\pi x)$